Essays

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There are different types of biomolecules in the human body. Each type has specific functions in the human body. Describe how the structures of proteins and nucleic acids are related to their functions. (11 marks)

##

Proteins (Max. 4)

- Proteins are long chains of polypeptides made up of amino acids joined by peptide bonds. (1)
- The human body metabolizes proteins to release energy when lipids and carbohydrates are in shortage. (1)
- Amino acids are deaminated in the liver. The remaining parts of amino acids are changed into carbohydrates and lipids, which can be respired to release energy. (1)
- Protein molecules have specific 3D structure (conformation). (1)
- Enzymes are protein molecules which have a specific active site to bind to substrate molecules. (1)
- Enzymes can speed up chemical reactions. (1)
- Antibodies are protein molecules which have specific binding site to the antigens on pathogens. (1)
- Antibodies can produce immune response against pathogens. (1)
- Some hormones are protein molecules which have structures specific to the receptors on target cells. (1)
- Hormones help to regulate different processes in the body. (1)

Nucleic acids (Max. 4)

- DNA and RNA are nucleic acids found in the human body. They consist of numerous nucleotides joined together. (1)
- The base sequences in DNA molecules carry genetic information. (1)
- The double helix structure stabilizes DNA. (1)
- RNA is involved in protein synthesis. (1)
- The base sequence in RNA determines the amino acid sequence in a polypeptide chain. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

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Discuss the significance of condensation and hydrolysis reactions to living organisms using carbohydrates as an example. (12 marks)

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- The condensation of monosaccharides forms disaccharides or polysaccharides. Water is produced in the process. (1)
- In hydrolysis, complex molecules are broken down into simpler molecules. Water is utilized in the process. (1)
- The condensation of glucose and fructose produces sucrose which is the transport form of sugars in plants. (1)
- The condensation of numerous glucose molecules produces cellulose, which is a major component of the cell wall in plant cells. (1)
- The condensation of numerous glucose molecules produces starch, which acts as an energy reserve in plants. (1)
- The condensation of numerous glucose molecules produces glycogen, which acts as an energy reserve in animals. (1)
- Starch and glycogen can be hydrolysed to release glucose molecules to be used in respiration to release energy. (1)
- When the blood glucose level is lower than normal, the glycogen stored in the liver can be hydrolysed to release glucose molecules to restore a normal blood glucose level. (1)
- Disaccharides and polysaccharides in ingested food are hydrolysed into monosaccharides to be absorbed through the intestinal walls. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Applying concepts	9
Communication	3

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Both triglycerides and phospholipids are important lipids in the human body. Describe the structural similarities and differences between triglycerides and phospholipids, and explain their functions in the human body with appropriate examples. (11 marks)

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Structural similarities and differences between triglycerides and phospholipids:

- Both a triglyceride molecule and a phospholipid molecule contain a glycerol molecule. (1)
- A triglyceride molecule consists of three fatty acid molecules. (1)
- A phospholipid molecule consists of two fatty acid molecules and a phosphate group. (1)

Functions of triglycerides and phospholipids in the human body:

- Triglycerides can be stored as energy reserves. (1)
- Adipose tissue under the skin (i.e. subcutaneous fat) acts as an insulating layer to reduce heat loss. (1)
- Adipose tissues around internal organs act as shock-absorbers for protection. (1)
- Phospholipids are the main components of cell membranes (1)
- which controls the entry and exit of molecules into and out of the cell. (1)

Communication (Max. 3)

(11 marks)

##

Skills assessed	Marks
Applying concepts	8
Communication	3

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